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MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING/

SCHOOL OF TECHNOLOGY MANAGEMENT

Academic Year: 2024-2025

Program/s: B.Tech /MBA Tech/BTI | Year: II/IV Semester: III/VII

Stream/s : Data Science

Subject: Managing Uncertainty | Time: 3 hrs (10.00 a.m to 1.00 p.m)

Date: 11/12/24 | No. of Pages:5

Marks: 100

Final Examination (2024-25) /Re-Examination (2022-23, 23-24)

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. _1___ is compulsory.
2) Out of remaining questions, attempt any __4___ questions.
3) **In all ___5_ questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

<table>
<tr><td>Q1</td><td></td><td>Answer briefly:</td><td>[20]</td></tr>
<tr><td>CO-3 ;
SO-2 ;
BL-2</td><td>a.</td><td>Explain different types of classification.</td><td>[5]</td></tr>
<tr><td>CO-3 ;
SO-2 ;
BL-3</td><td>b.</td><td>The following data relating to the strength of the Indian Merchant Shipping Fleet gives the Gross Registered Tonnage (GRT) as on 31st December, for different years.
<table>
<tr><td>Year</td><td>1961</td><td>1966</td><td>1971</td><td>1975</td><td>1976</td></tr>
<tr><td>GRT in '000</td><td>901</td><td>1792</td><td>2500</td><td>4464</td><td>5115</td></tr>
</table>
Construct the data by suitable bar diagram.</td><td>[5]</td></tr>
<tr><td>CO-1 ;
SO-2 ;
BL-1</td><td>c.</td><td>In a single throw with two uniform dice find the probability of throwing (i) Five, (ii) Eight.</td><td>[5]</td></tr>
</table>

CO-2 ; SO-2 ; BL-2	d.	Explain PMF, PDF and CDF. Also, explain the relation between them.	[5]
Q2.a CO-3; SO-2; BL-5		The following are the marks of 150 students in an examination. Evaluate Karl Pearson's coefficient of skewness. Comment on the answer	[10]

Marks	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80
No. of students	10	40	20	0	10	40	16	14

Q2.b CO-3; SO-2; BL-5

For the following frequency distribution, determine (a) The median class. (b) The number of the item that represents the median. (c) The width of the equal steps in the median class. (d) The estimated value of the median for these data. (e) Estimate the value of Median with the formula. [10]

Class	Frequency
100–149.5	12
150–199.5	14
200–249.5	27
250–299.5	58

Class	Frequency
300–349.5	72
350–399.5	63
400–449.5	36
450–499.5	18

Q3.a CO-1; SO-2; BL-5

A part is observed to have no defect. Estimate the probability that it was produced by X manufacturer? (ii) A part is known to have been produced by manufacturer X. Estimate the probability that the part has no defects? (iii) A part is known to have two or more defects. Estimate the probability that it was manufactured by X? (iv) A part is known to have one or more defects. Estimate the probability that it was manufactured by Y? [10]

Manufacturer	Number of Defects			
	0	1	2	3
X	0.1250	0.0625	0.1875	0.1250
Y	0.0625	0.0625	0.1250	0.2500

Q3.b CO-1; SO-2; BL-5		A company launches an advertising campaign of its new product on TV, radio and in print media in an area where 30% watch TV, 50% listen to the radio and the rest rely on newspapers for all information. It is estimated that a person who sees the advertisement on TV will buy the product with probability 0·6. A person who has heard it on radio is expected to buy the product with probability 0·3 and seeing the advertisement in print will convince a person to buy the product with probability 0·1. A consumer, chosen at random, is found to have purchased the product. Evaluate the probability she heard about the product on radio?	[10]
Q4.a CO-2; SO-2; BL-5		The random variable X has binomial distribution B (20, 0.4). Determine each of the following. a) P (X = 6). b) P (X ≤ 9). c) P (X ≥ 12). d) P (8 ≤ X≤ 13).	[10]
Q4.b CO-2; SO-2; BL-5		A discrete random variable has the probability density function given below. Evaluate k, mean and variance X: -2, -1, 0, 1, 2, 3 P(X=x): 0.2, k, 0.1, 2k, 0.1, 2k	[10]
Q5.a CO-2; SO-2; BL-4		It is given that 2% of the screws are defective. Apply Poisson Distribution to determine the probability that a packet of 100 screws contains: (1) No defective screws (2) One defective screw (3) Two or more defective screw (4) Two or less defective screw	[10]
Q5.b CO-2; SO-2; BL-4		Suppose that a trainee soldier shoots a target in an independent fashion. If the probability that the target is hit in any shot is 0.7. Determine the (i) Probability that the target would hit on the 10th attempt? (ii) probability that it taken him less than 4 shots? (iii) Probability that it taken him an even number of shots? (iv) Average number of shots needed to hit the target.	[10]
Q6.a CO-1; SO-2; BL-4		The joint density function of X and Y is given by P(x,y) = k(2x+y); x = 0,1,2; y = 0,1,2 Distinguish (i) value of k (ii) Marginal distribution of X and Y (iii) are X and Y independent? (iv) find the conditional distribution of Y for X = x.	[10]

Q6.b CO-3; SO-2; BL-3		Two independent random samples of series are drawn from normal population with the same variance. The sample results are given below: $n_1 = 4, \bar{x} = 12, s_1 = 4$ $n_2 = 5, \bar{y} = 10, s_2 = 3$ where s_1^2 and s_2^2 are unbiased estimate of the common population variance. Also identify 90% confidence interval for $\mu 1 - \mu 2$	[10]
Q7.a CO-3; SO-2; BL-4		Based on field experiments, a new variety green gram is expected to give an yield of 12.0quintals per hectare. The variety was tested on 10 randomly selected farmers' fields. The yield (quintals/hectare) were recorded as 14.3,12.6,13.7,10.9,13.7,12.0,11.4,12.0,12.6,13.1. Do the results conform the expectation?	[10]
Q7.b CO-1; SO-2; BL-2		State Bayes theorem. Explain in detail with proof	[10]

t Table

cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
one-tail	0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
two-tails	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
z	0.000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291
	0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
	Confidence Level										